# Counting stable tie refinements

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There are equally many agents on two sides. Each agent gives a complete weak preference order, and $M$ is a specified perfect matching. Break each tie uniformly and independently into a strict order. An unmatched pair blocks $M$ when both agents prefer each other to their partners. This is the compact-indifference model of [1, Section 3.1.2, Table 1] and [2, Sections 2 and 5, Table 3]. If the tie sizes are $t_{a,j}$, set

$$D(I) = \prod_{a,j} t_{a,j}!, \qquad N(I, M) = \#\{\text{strict refinement profiles in which } M \text{ is stable}\}.$$

Thus $p_I(M) = N(I, M)/D(I)$. The weak orders are explicit and any rational threshold is encoded in binary.

**Theorem 1.** *The integer function $N$ is $\#\mathsf{P}$-complete under polynomial-time Turing reductions. Computing the exact rational probability $p$ and computing $N$ are polynomial-time Turing equivalent. The language*

$$\{(I, M, \tau) : \ \tau \in [0, 1] \text{ is rational and } p_I(M) \geq \tau\}$$

*is $\mathsf{PP}$-complete under polynomial-time Turing reductions. No complexity separation is assumed. Here Turing reductions allow polynomially many adaptive oracle queries.*

Counting perfect matchings of a simple balanced bipartite graph is equivalent to computing the permanent of a 0–1 matrix, which is $\#\mathsf{P}$-complete [3, Theorem 1]. Section 2 of that paper uses polynomial-time oracle reductions.

For a simple bipartite graph $H = (V, E)$ of minimum degree at least two, write $h = |V|$, $e = |E|$, and $d_v = \deg_H(v)$. Let $C(H)$ be the number of ways for every vertex to choose one incident edge such that no edge is chosen by both endpoints.

**Lemma 1.** *$C(H)$ can be computed in polynomial time with an exact compact-indifference stability-probability oracle.*

*Proof.* If $H$ is empty, $C(H) = 1$. For nonempty $H$ and an integer $t \geq \Delta(H)$, give each original vertex $v$ a unique private partner on the opposite side. Add $r_v = t - d_v$ distinct probe agents on the opposite side, each with its own private partner. The matching $M$ consists of these private pairs. Every original vertex has one top tie containing its partner, its $H$-neighbors, and its probes; it has size $t + 1$. A probe of $v$ strictly ranks $v$ first, its own partner second, and all others below. Every private partner ranks its matched agent first. Put all remaining names in a fixed strict order below the partner. Every private pair adds one agent to each side, so the sides are balanced.

Only edges of $H$ and original-vertex–probe pairs can block: every other unmatched pair has an endpoint that ranks its partner above the other agent. Generate each top-tie permutation using independent uniform priorities in $[0, 1]$, smaller first. Write $x_v$ for its partner's priority. Conditional on all $x_v$, a given tied competitor precedes the partner with probability $x_v$. Different unmatched pairs use distinct competitor priorities, so their safety events are conditionally independent. A probe pair is safe with probability $1 - x_v$; an edge $vw$ of $H$ is safe with probability $1 - x_v x_w$. Therefore the oracle gives

$$p_H(t) = \int_{[0,1]^h} \prod_v (1 - x_v)^{t - d_v} \prod_{vw \in E} (1 - x_v x_w)\, dx. \tag{1}$$

The $h$ top ties are the only nontrivial ties, giving $D = ((t+1)!)^h$ refinement profiles.

For $F \subseteq E$, let $k_v = \deg_F(v)$. Expanding the finite edge product and using

$$\int_0^1 x^k (1-x)^r \, dx = \frac{k!\, r!}{(r+k+1)!}$$

gives, for the nonnegative integers $r = t - d_v$,

$$Q(t) := p_H(t) \prod_v (t - d_v + 1) = \sum_{F \subseteq E} (-1)^{|F|} \prod_v L_{t-d_v}(k_v), \qquad L_r(k) = \frac{k!}{\prod_{j=2}^{k+1}(r+j)}. \tag{2}$$

The empty denominator product is one. The integral identity follows by $k$ integrations by parts, starting with $\int_0^1 (1-x)^r dx = 1/(r+1)$. Set

$$P(t) = \prod_v P_v(t), \qquad P_v(t) = \prod_{j=2}^{d_v+1} (t - d_v + j), \qquad K = \sum_v d_v = 2e.$$

Multiplying each term in (2) by $P$ cancels all denominators:

$$A(t) := P(t)Q(t) = \sum_{F \subseteq E} (-1)^{|F|} \prod_v \left[ k_v! \prod_{j=k_v+2}^{d_v+1} (t - d_v + j) \right]. \tag{3}$$

Thus $A$ is a polynomial of degree at most $K$. Query (1) at the $K+1$ valid integers $t = \Delta(H), \ldots, \Delta(H)+K$, calculate $A(t)$, and interpolate its value at zero. All queries use nonnegative probe counts; evaluation at zero takes place only in polynomial (3).

Since $d_v \geq 2$, the vertex factor in (3) at zero is

$$k! \prod_{j=k+2}^{d_v+1} (j - d_v) = \begin{cases} 0, & k \leq d_v - 2, \\ (d_v-1)!, & k = d_v - 1, \\ d_v!, & k = d_v. \end{cases} = (d_v-1)! \binom{k}{d_v - 1}.$$

Consequently

$$\frac{A(0)}{\prod_v (d_v-1)!} = \sum_{F \subseteq E} (-1)^{|F|} \prod_v \binom{\deg_F(v)}{d_v - 1}. \tag{4}$$

For each $v$, the binomial factor counts a selection $B_v$ of $d_v - 1$ incident edges contained in $F$. Exchange the finite sums: a fixed family $(B_v)$ contributes $\sum_{F \supseteq B} (-1)^{|F|}$, where $B = \bigcup_v B_v$. This sum is zero if $B \neq E$, by pairing subsets according to any edge of $E \setminus B$; it equals $(-1)^e$ if $B = E$. Selecting $B_v$ omits exactly one incident edge at $v$. The union is $E$ precisely when no edge is omitted at both endpoints. The omitted edges therefore give precisely the choices counted by $C(H)$:

$$C(H) = \frac{(-1)^e A(0)}{\prod_v (d_v-1)!}. \tag{5}$$

Put $T = \Delta + K$. Each query uses $O(h(T+1))$ agents with explicit complete lists, and its profile denominator has $O(hT \log(T+1))$ bits. At a query point, $0 \leq p_H(t) \leq 1$ and all $K + h$ factors in $P(t) \prod_v (t - d_v + 1)$ are at most $T + 1$; hence the integer $A(t)$ has at most $O((K+h)\log(T+1))$ bits. Each Lagrange coefficient at zero uses $K$ numerators of magnitude at most $T$ and $K$ nonzero denominator differences of magnitude at most $K$. A common denominator over the $K+1$ summands therefore has $O(K^2 \log(T+1))$ bits. All arithmetic is consequently polynomial. □

**Lemma 2.** *The number of perfect matchings of any simple balanced bipartite graph can be computed in polynomial time with an oracle for $C(H)$ on simple bipartite graphs of minimum degree at least two.*

*Proof.* Let $G$ be simple and balanced bipartite, with $V = 2n > 0$ vertices and degrees $d_v$. The empty graph on zero vertices has one perfect matching and needs no oracle. For each integer $R = \Delta(G) + 1, \ldots, \Delta(G) + V + 1$, attach $\ell_v = 3(R - d_v)$ disjoint four-cycles to $v$, using one bridge from $v$ to a designated vertex $z$ of each cycle. Call the resulting graph $H_R$ and write $L = \sum_v \ell_v$. Each cycle can be bipartitioned with $z$ opposite $v$, so $H_R$ is bipartite and simple. The original vertices have degree at least three, the designated cycle vertices have degree three, and the others degree two.

Consider one cycle $z - a - b - c - z$ and its bridge to $v$. If $v$ chooses the bridge, $z$ cannot choose it. The four cycle vertices must then choose cycle edges with no mutual choice. Following choices reaches a directed cycle; with mutual choices excluded it must use all four vertices. There are two orientations. If $v$ does not choose the bridge, those same two choices remain, and there are four choices with $z$ choosing $v$. For the latter count, $b$ chooses either $a$ or $c$; its chosen neighbor must choose $z$, while the other neighbor has two choices. These two cases give four.

Factor out six choices per cycle. At an original vertex, choosing an original edge has weight one, and choosing one of its bridges has weight $2/6 = 1/3$. Its total choice weight is therefore $d_v + \ell_v/3 = R$. Apply inclusion-exclusion to the remaining forbidden events, in which both endpoints choose an original edge. An intersection of events sharing an endpoint is impossible, since a vertex chooses only one edge. The nonzero intersections therefore correspond to matchings $F$ of $G$. Each fixes choices at $2|F|$ vertices, with weight one, and leaves total weight $R$ at every other vertex. We obtain

$$Z_G(R) := \frac{C(H_R)}{6^L} = \sum_{\substack{F \subseteq E(G) \\ F \text{ matching}}} (-1)^{|F|} R^{V-2|F|}. \tag{6}$$

The right side is a polynomial of degree $V$. Its constant coefficient is $(-1)^n \operatorname{PM}(G)$, since the only matchings with $V - 2|F| = 0$ are perfect matchings. Interpolating at the $V + 1$ chosen $R$ values therefore recovers $\operatorname{PM}(G)$.

Here $R = O(V)$, $L = O(V^2)$, and $H_R$ has $O(V^2)$ vertices and edges. The factor $6^L$ has polynomial bit length, and $0 \le Z_G(R) \le R^V$ by the weighted-choice interpretation. Exact interpolation has the same polynomial bit bound as above. Combining with Lemma 1, the largest matching instance has $O(V^4)$ agents and $O(V^8)$ preference-list entries, and the total number of queries is polynomial. □

*Proof of Theorem 1.* Encode every tie permutation by a fixed-length list of its element indices, rejecting invalid lists. Every complete strict refinement has exactly one valid encoding. A nondeterministic machine guesses the concatenation, checks validity, and checks all unmatched pairs for blocking in polynomial time. Its accepting-branch count is exactly $N$, so $N \in \#\mathsf{P}$.

The product $D$ is computable in polynomial time and has polynomial bit length. Given $N$, compute $p = N/D$; given exact $p$, compute the integer $N = Dp$. Thus the functions are Turing equivalent. Lemmas 1–2 and Valiant's theorem prove $\#\mathsf{P}$-hardness and hence the stated completeness of $N$.

For threshold membership, write $\tau = a/b$ with $b > 0$ and set $k = \lceil aD/b \rceil$. Then $p \ge \tau$ is equivalent to $N \ge k$. Let $B$ be the length of the fixed encoding just used, so the counting machine has $2^B$ branches and $N \le D \le 2^B$. If $k = 0$, accept deterministically. Otherwise construct a machine with $2^{B+1}$ branches by guessing a leading bit and then $B$ further bits. In its first half it performs the counting test, giving $N$ accepting branches. In its second half it interprets the $B$ bits as an integer and accepts exactly the first $2^B - k + 1$ integers. Its accepting count is $N + 2^B - k + 1$, strictly above half of all branches exactly when $N \ge k$. The bound $1 \le k \le D \le 2^B$ makes that second-half count valid. The machine runs in polynomial time, proving membership in $\mathsf{PP}$.

A threshold oracle recovers $N$ by binary search over $0, \ldots, D$, using thresholds $k/D$ and $O(\log(D+1))$ calls. Thus the threshold oracle computes every #P function. For any language in PP, take a polynomial-time majority machine making a fixed $q(|x|) \geq 1$ binary guesses. Its accepting count is a #P function; test whether it exceeds $2^{q(|x|)-1}$. This gives a polynomial-time Turing reduction from every PP language to the threshold problem and proves the claimed PP-hardness. □

## Use of AI tools

The author used research AI agents that the author built and configured (GPT-6 Astra max, Claude Opus 5.5 max and GPT-6 Astra Pro) during the proof work, for proof search and construction, proof checking, translation, Python computation and similar tasks. The final manuscript was written, and the proofs were checked, by the human author. All responsibility for this work rests with the author personally.

## References


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